Image to Animation

Image to Animation
1. Function Introduction
2. Hardware Connection and Setup
3. Code Analysis
4. Running the Code
5. Experimental Results

1. Function Introduction

This example program can generate a 5-second animation from a single image.

2. Hardware Connection and Setup

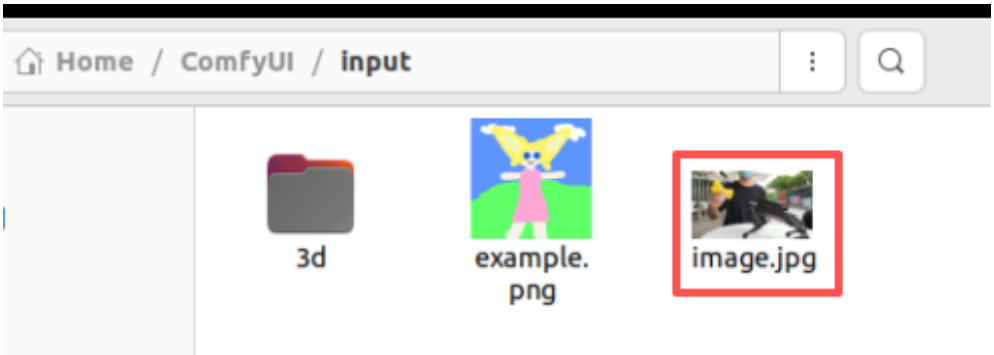
This example program requires not only a power adapter but also a DP monitor, mouse, and keyboard for displaying the interface and operation.

3. Code Analysis

Source code path:

```
/home/jetson/showing/animation/animation.py
```

If you need to replace the image, please replace the image in the path ~/ComfyUI/input/image.jpg.



If you need to modify the prompt message, you can modify the PROMPT variable.

```
TEXT_PROMPT = '''The background is a cyberpunk-style Tang Dynasty Never Sleeping
City, with flying eaves and holographic lanterns. The camera slowly zooms in and
circles around. The image shows a robotic dog, and a person is slowly waving a
yellow bone in front of it. The robotic dog follows the bone's sway, and the
movements are natural and seamless, with a smooth transition between real and
virtual elements. \nConstraint:\nDo not change the robotic dog's appearance.'''
```

If you need to change the output file name, modify the OUTPUT_NAME variable. The final output will be in the ~/ComfyUI/output/video/ directory.

```
OUTPUT_NAME = "myvideo"
```

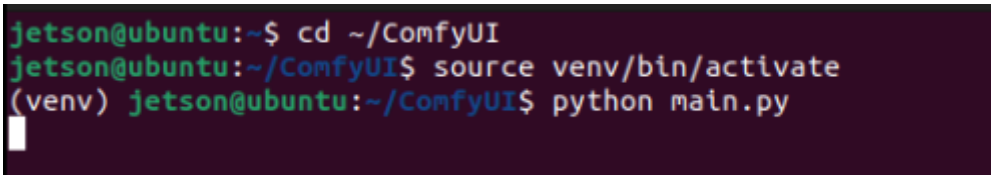
If you need to change the ComfyUI port, modify the value of the self.comfyui_server variable.

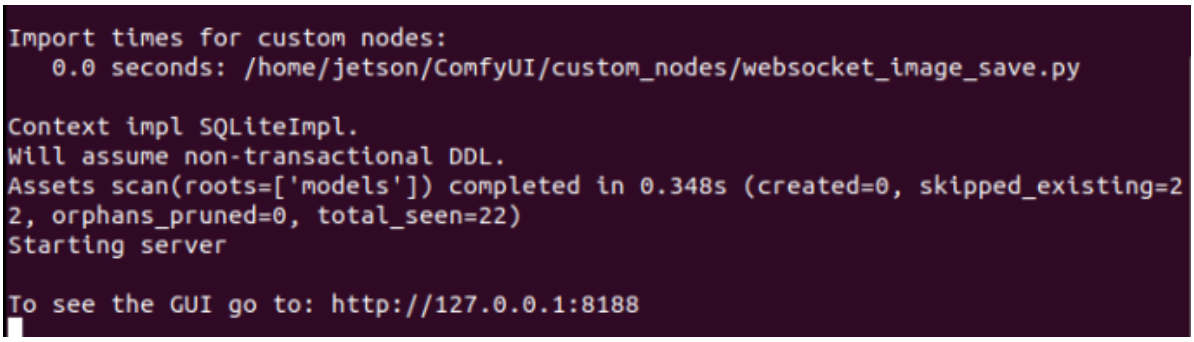
```
self.comfyui_server = '127.0.0.1:8188'
```

4. Running the Code

Open a new system terminal and start the ComfyUI service.

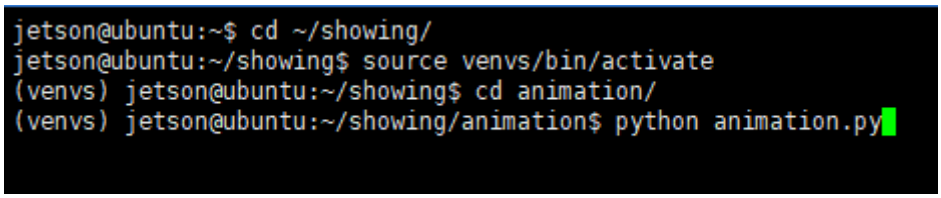
```
cd ~/ComfyUI
source venv/bin/activate
python main.py
```





Open another new system terminal, start the virtual environment, and start the main program for telling stories.

```
cd ~/showing/
source venvs/bin/activate
cd animation/
python animation.py
```



5. Experimental Results

After the program starts, it will call a large model file to render the animation. The program will take a relatively long time to run; the default configuration takes about 16 minutes.

```
jetson@ubuntu: ~/ComfyUI
bj.bias']
VAE load device: cuda:0, offload device: cpu, dtype: torch.float16
Requested to load CLIPVisionModelProjection
Loaded completely; 52354.41 MB usable, 787.72 MB loaded, full load: True
Found quantization metadata version 1
Using MixedPrecisionOps for text encoder
Requested to load QwenImageTEModel_
Loaded completely;  8328.59 MB loaded, full load: True
CLIP/text encoder model load device: cuda:0, offload device: cpu, current: cuda:
0, dtype: torch.float16
Requested to load AutoencodingEngine
Loaded completely; 35603.63 MB usable, 2408.48 MB loaded, full load: True
model weight dtype torch.float16, manual cast: None
model_type FLOW
EasyCache enabled - threshold: 0.2, start_percent: 0.15, end_percent: 0.95
Requested to load HunyuanVideo15
Loaded completely; 24005.73 MB usable, 15881.76 MB loaded, full load: True
  0%|                                                    | 0/20 [00:00<?, ?it/s]
Interrupting prompt e636ec9f-87e1-4160-b0d0-ff492f90f620
  0%|                                                    | 0/20 [00:30<?, ?it/s]
EasyCache - skipped 0/20 steps (1.00x speedup).
Processing interrupted
Prompt executed in 65.73 seconds
```

```
jetson@ubuntu: ~/showing/animation
==================================================
The program has been started.
Please open your browser and log in to the ComfyUI backend (http://127.0.0.1:818
8) to check the progress.
After the video is completed, it will be automatically saved in the ComfyUI/outp
ut/video/ folder.

(venvs) jetson@ubuntu:~/showing/animation$ python animation.py
Starting Interactive Animation...
[ANIM] Generating: 背景是赛博朋克大唐不夜城，飞檐斗拱配全息灯笼，镜头慢推环绕，
图片中是一只机器狗，有一个人拿着黄色的骨头在机器狗面前缓慢摇晃，机器狗跟着黄色骨
头摇晃，机器狗动作衔接自然无卡顿，画面虚实结合，过渡丝滑无断层。
限制条件：
不要改变机器狗的样貌


==================================================
The program has been started.
Please open your browser and log in to the ComfyUI backend (http://127.0.0.1:818
8) to check the progress.
After the video is completed, it will be automatically saved in the ComfyUI/outp
ut/video/ folder.

(venvs) jetson@ubuntu:~/showing/animation$
```

When the progress bar reaches 100%, it indicates completion.

```
Prompt executed in 65.73 seconds
got prompt
EasyCache enabled - threshold: 0.2, start_percent: 0.15, end_percent: 0.95
100%|[redacted]| 20/20 [16:03<00:00, 48.19s/it]
EasyCache - skipped 9/20 steps (1.82x speedup).
Prompt executed in 00:17:22
```

After the animation is generated, the MP4 video file is stored in the ~/ComfyUI/output/video/ directory.

```
jetson@ubuntu:~$ cd ComfyUI/output/video/
jetson@ubuntu:~/ComfyUI/output/video$ ls
myvideo_00001_.mp4
jetson@ubuntu:~/ComfyUI/output/video$
```